

What we're doing

A small, interactive, runnable check of how the 1B MarFold model predicts CA-CA distance maps on 10 random held-out AFDB structures from the `contacts-and-distances-v1-5x` test split. Anyone training a new checkpoint can drop it in and see how it stacks up.

Four variants run on the same 10 proteins: V1 zero-shot heatmaps; V2 beam-2 search over long-range GT contacts; V3 pure-greedy search over all three contact ranges; V4 directed search by predicted CB-CB distance (no per-candidate inference, LDDT-CB only).

Why

Existing eval surfaces are either the training-time tagged loss on 13 fixed PDB targets, or a one-PDB offline distogram script. This is the smallest useful "did the model learn the test distribution?" check. Hypothesis: zero-shot CA-CA MAE comparable to FoldBench ($\sim 3\text{-}4$ Å); heatmaps look like noisy versions of GT — diagonal correct, off-diagonal placed roughly, detail degrades at long range.

Results — zero-shot (V1)

****Macro CA-CA MAE = 3.29 Å**** across 10 AFDB proteins (40,442 pairs), on RTX A5000, ~5 min end-to-end. Per-protein MAE ranges 1.03 → 4.81 Å. Source CSV `data/per\_protein\_mae.csv`; per-protein heatmaps in `plots/`.

Results — seeded long-range only (V2)

Beam-2 search over GT long-range contacts ($CB-CB \leq 8 \text{ \AA}$, $sep \geq 24$), target $MAE < 1.0 \text{ \AA}$, $MAX_CONTACTS=30$. **2 / 10 proteins cross the target** ($k=7$ and $k=18$). The two crossers are the same two with the best zero-shot MAE. "Hardness" tracks number of GT long-range contacts.

Results — seeded all ranges (V3)

Pure greedy, candidate pool opened to long + medium + short. ****4 / 10** proteins cross the 1 Å target****** (vs 2 / 10 under V2), and ***every*** protein improves vs V2. Headline: letting the search pick from all three contact ranges matters more than the choice of search algorithm.

Results — directed search by predicted CB-CB (V4)

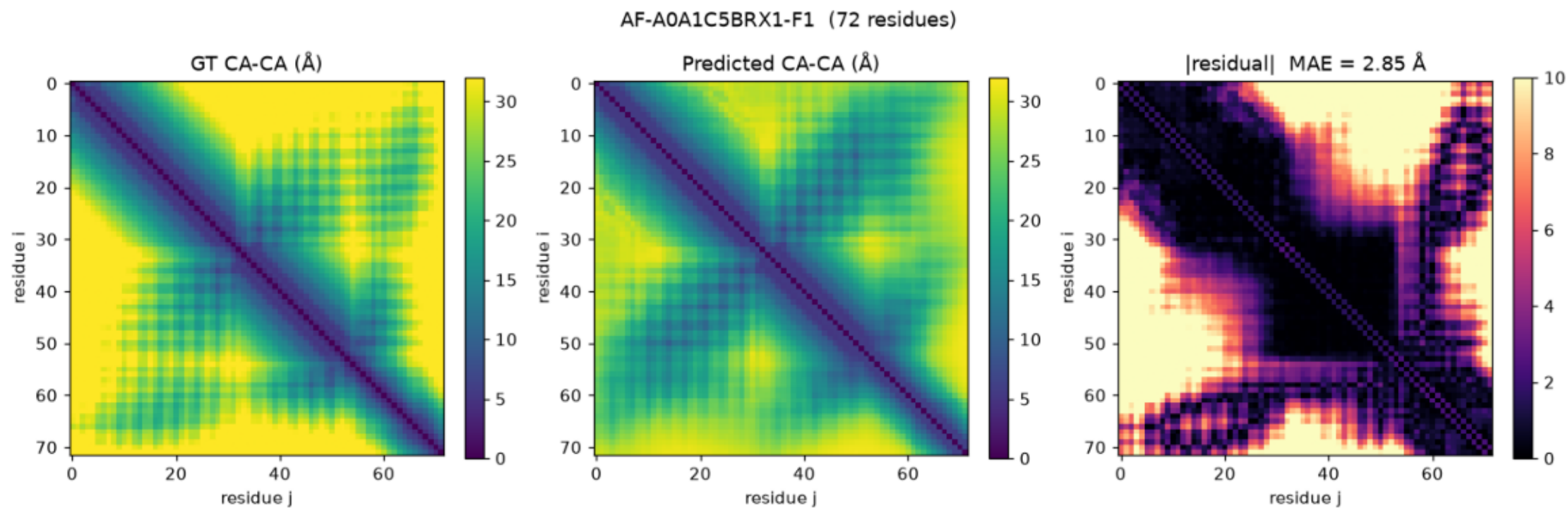
Single full-matrix prediction per round; pick next contact by sorting remaining candidates by current predicted CB-CB distance and taking the largest. Tracks LDDT-CB only (no MAE column). About $\frac{2}{3}$ the wall time of V3 + LDDT replay. Per-protein LDDT-CB curves in ``plots/contact_directed_search_trace.png``.

Conclusion

Zero-shot CA-CA MAE 3.29 Å is consistent with the ~3-4 Å hypothesis and close to FoldBench-monomer (~3.6 Å). With all-range seeding, 4 / 10 proteins cross 1 Å MAE; the hardest cases need either >30 contacts or a stronger model. The two notebooks are the deliverable — the future "did the new model learn?" check is: add a nickname to `MODELS.yaml` and re-run them.

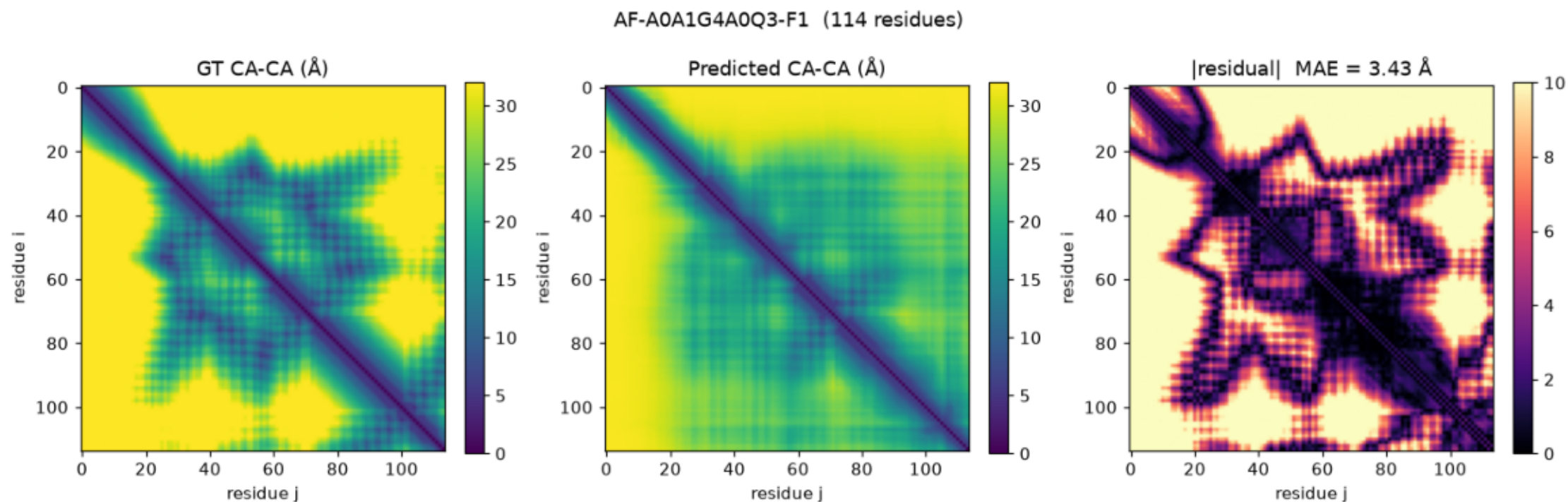
AF-A0A1C5BRX1-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-A0A1C5BRX1-F1 (72 aa, MAE 2.85 Å). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



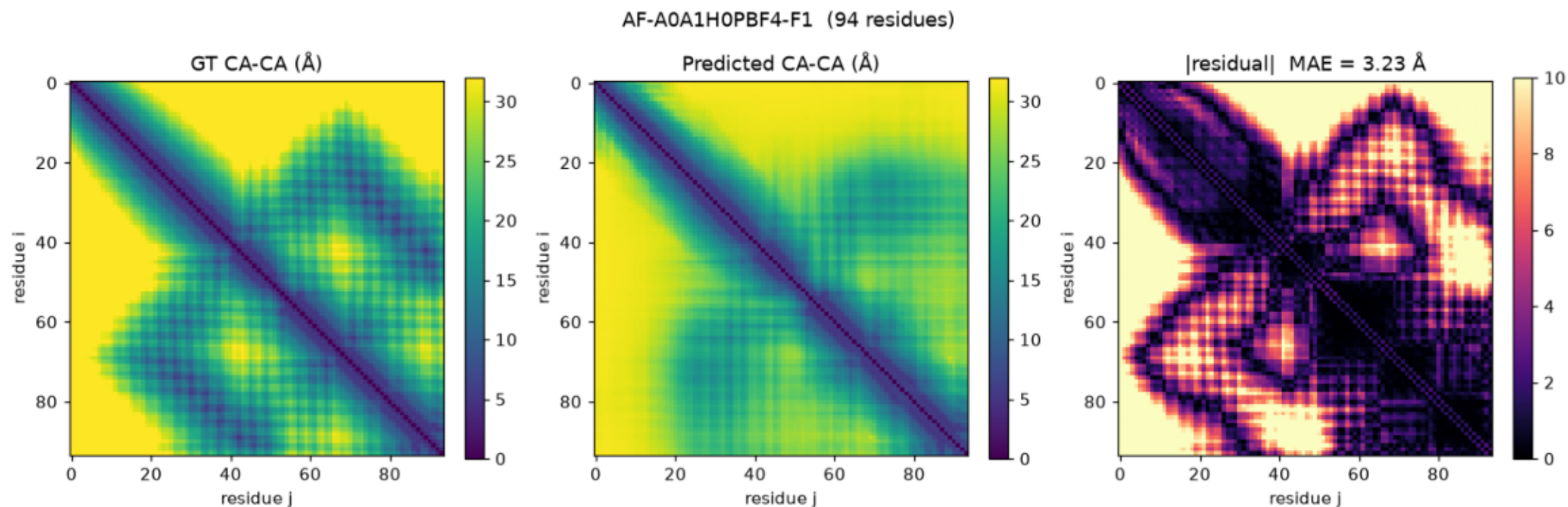
AF-A0A1G4A0Q3-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-A0A1G4A0Q3-F1 (114 aa, MAE 3.43 Å). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



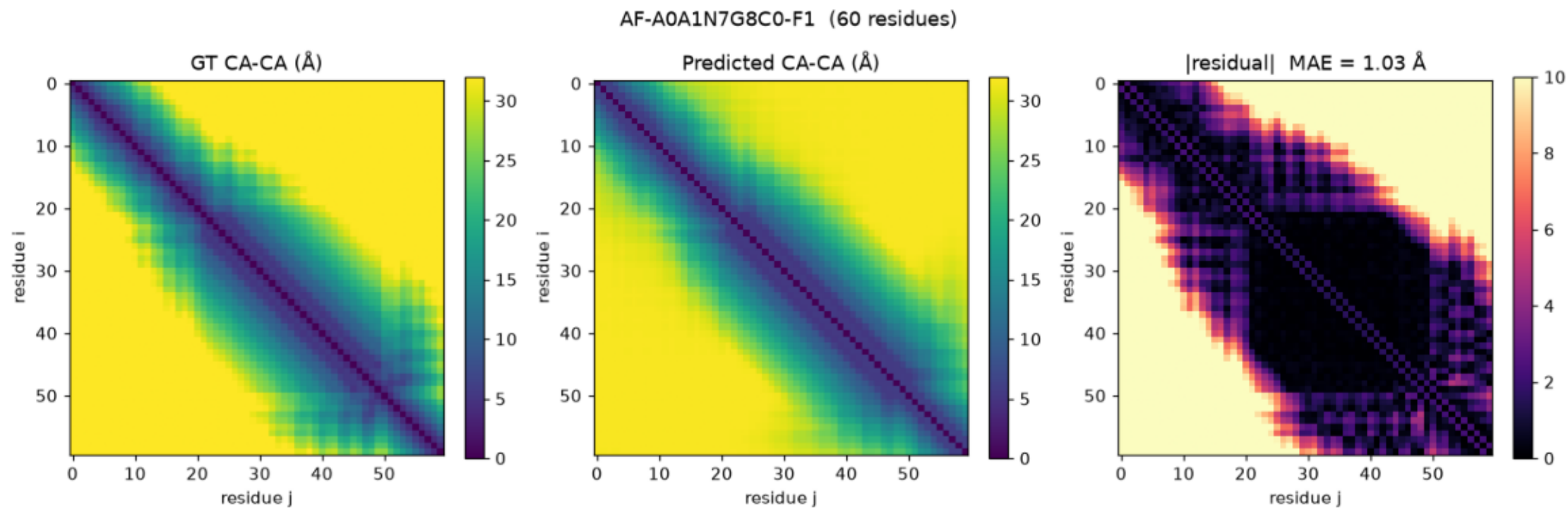
AF-A0A1H0PBF4-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-A0A1H0PBF4-F1 (94 aa, MAE 3.23 Å). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



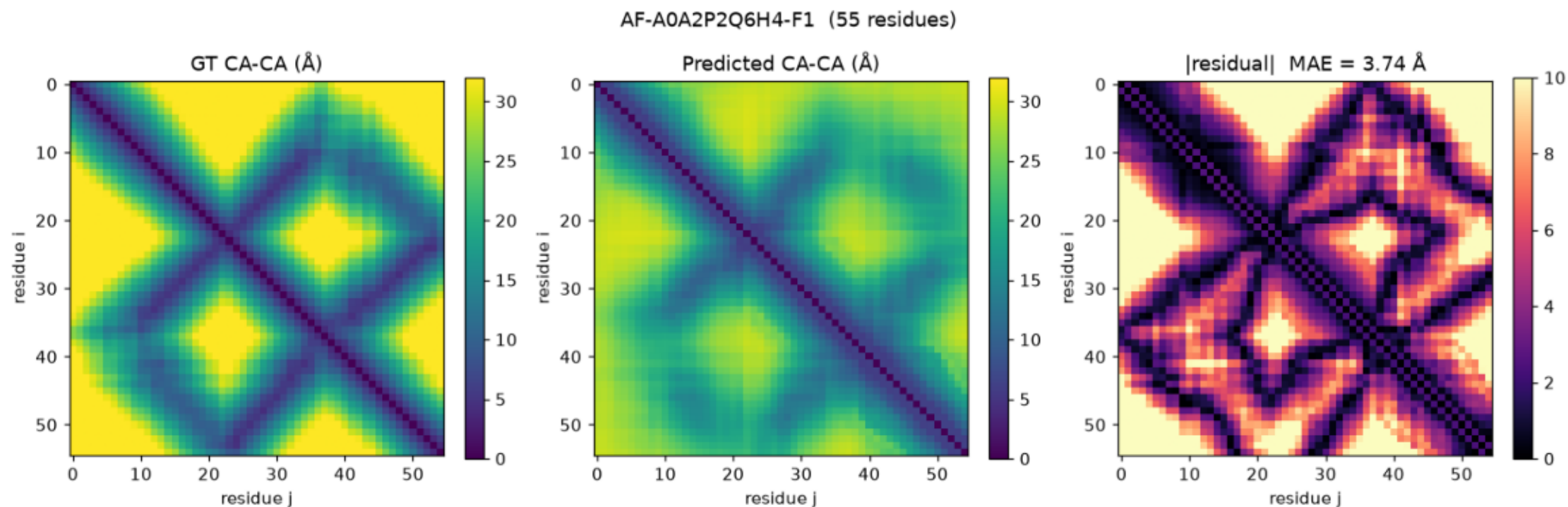
AF-A0A1N7G8C0-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-A0A1N7G8C0-F1 (60 aa, MAE 1.03 Å (best zero-shot)). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



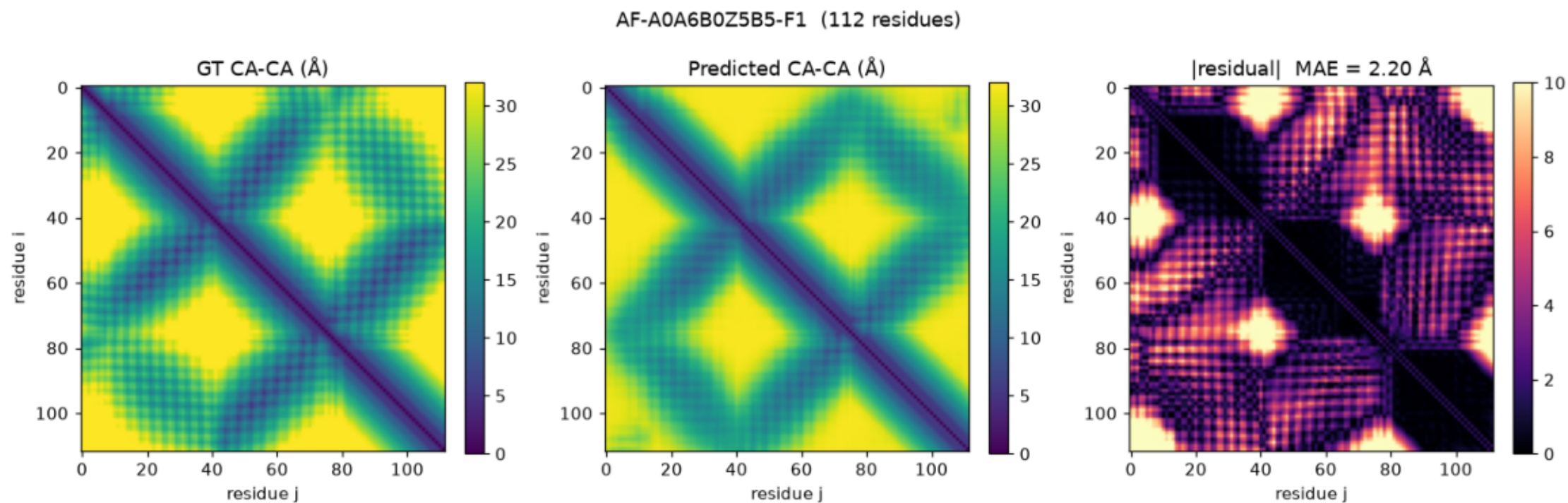
AF-A0A2P2Q6H4-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-A0A2P2Q6H4-F1 (55 aa, MAE 3.74 Å). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



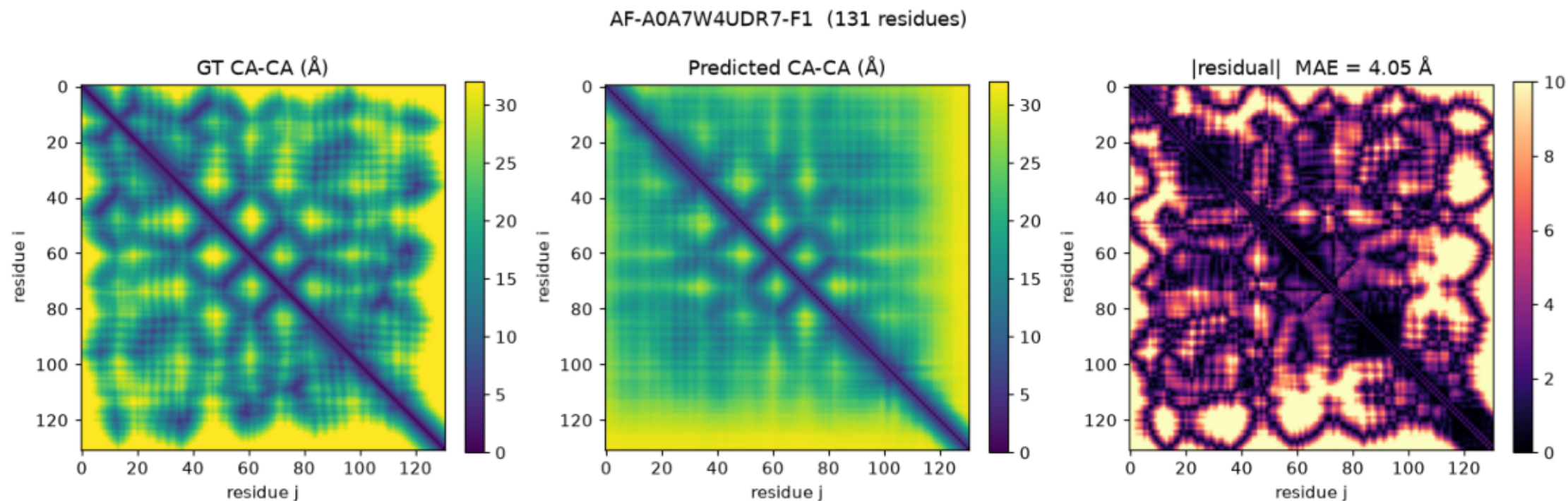
AF-A0A6B0Z5B5-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-A0A6B0Z5B5-F1 (112 aa, MAE 2.20 Å). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



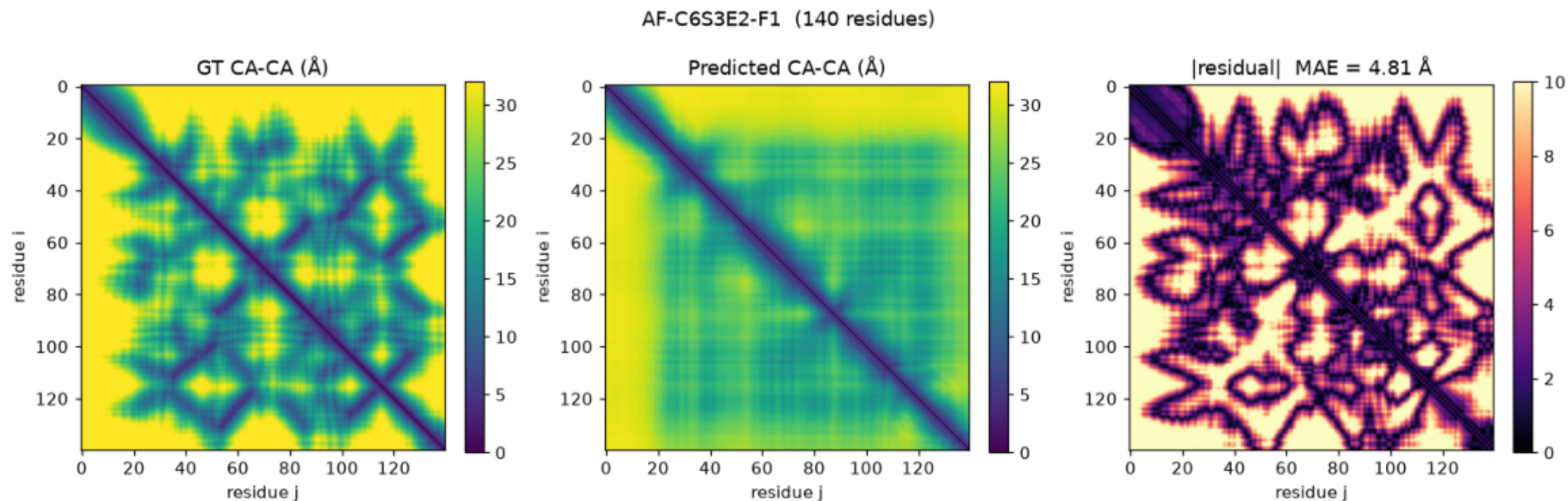
AF-A0A7W4UDR7-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-A0A7W4UDR7-F1 (131 aa, MAE 4.05 Å). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



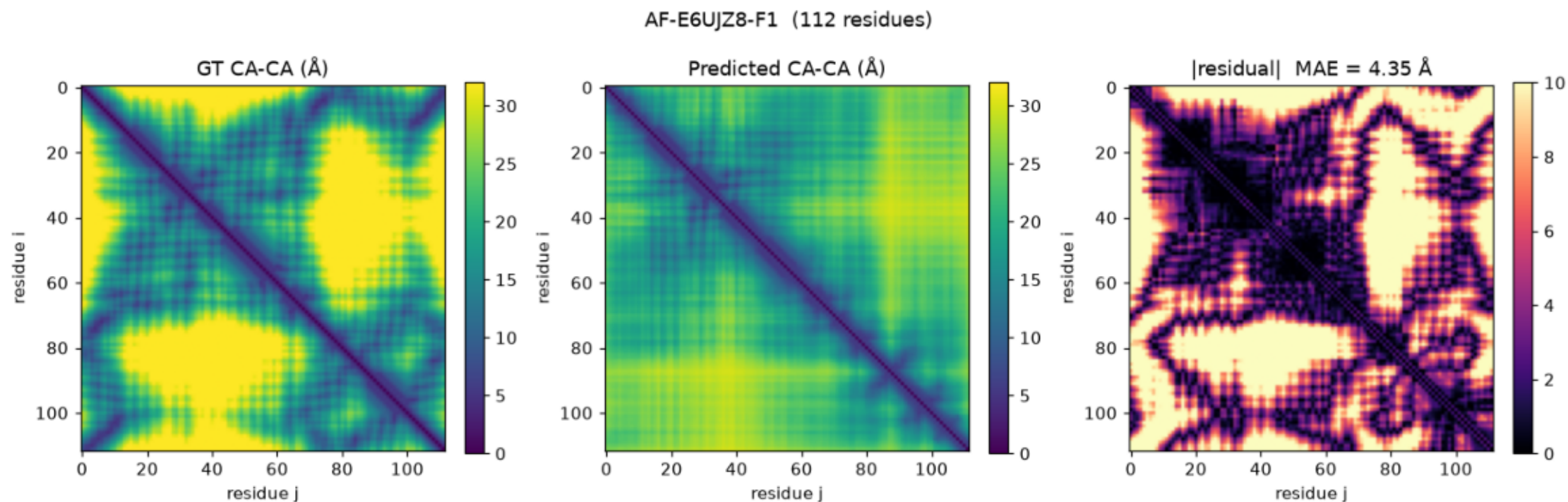
AF-C6S3E2-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-C6S3E2-F1 (140 aa, MAE 4.81 Å (hardest)). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



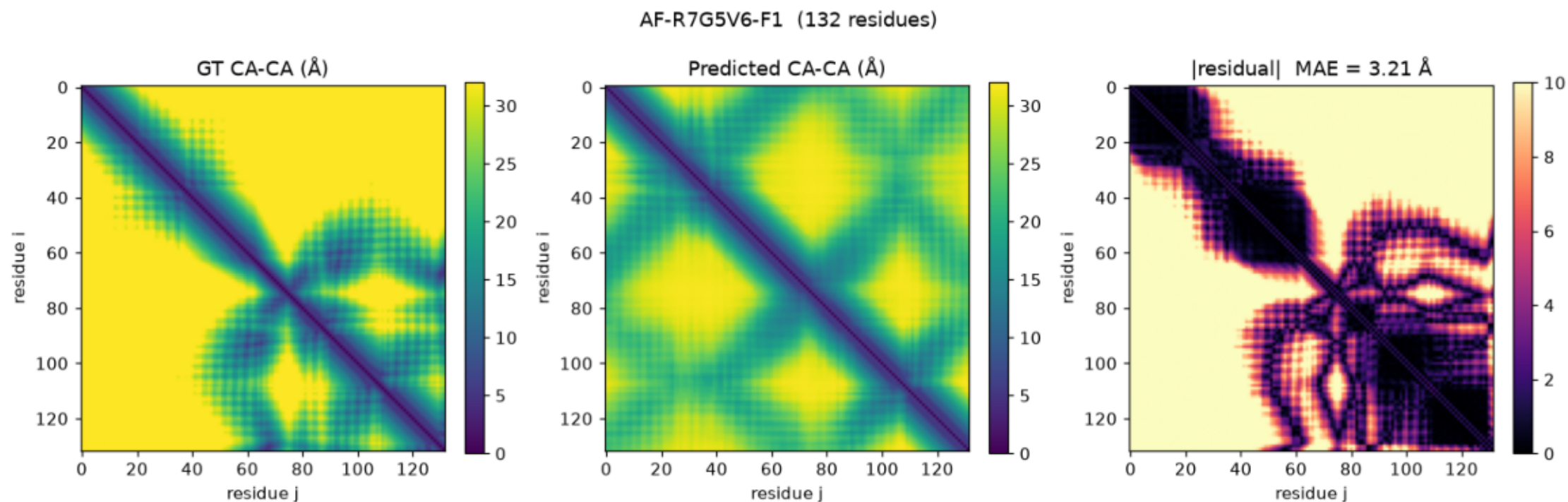
AF-E6UJZ8-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-E6UJZ8-F1 (112 aa, MAE 4.35 Å). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



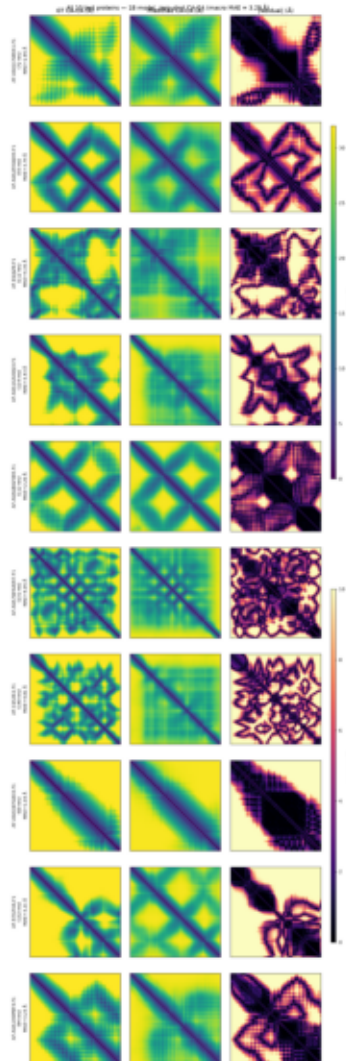
AF-R7G5V6-F1.png

V1 (zero-shot) — per-protein CA-CA distance heatmaps for AF-R7G5V6-F1 (132 aa, MAE 3.21 Å). Panels: GT, predicted, |residual|. Model: 1B, RTX A5000.



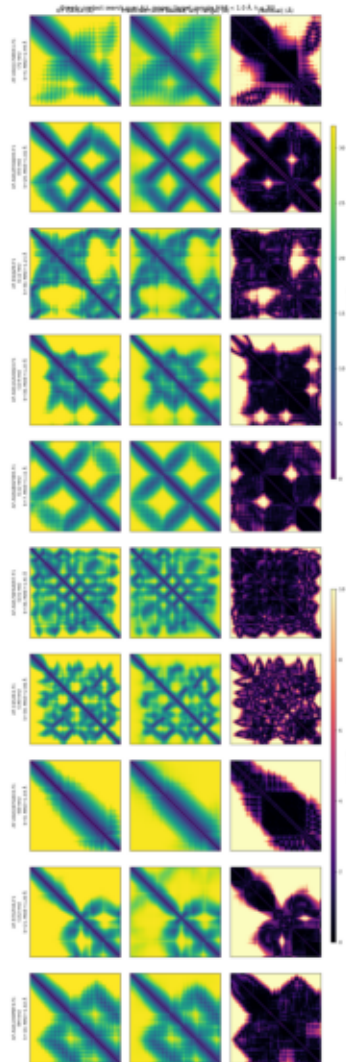
all_proteins_grid.png

V1 (zero-shot) — 10×3 grid of CA-CA heatmaps (GT / predicted / |residual|) across all 10 AFDB test proteins. Macro MAE = 3.29 Å.



contact_search_all_ranges_grid.png

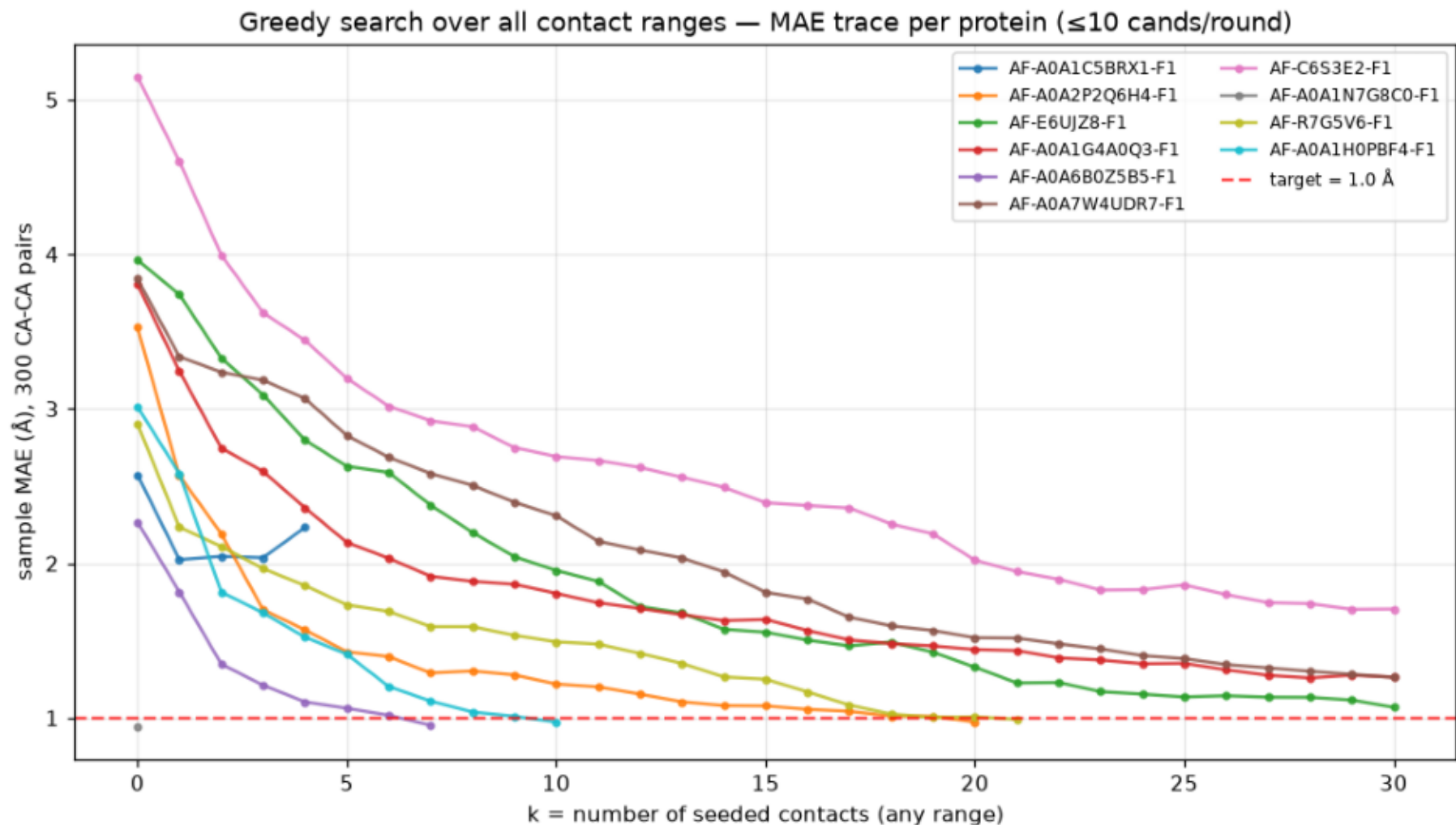
V3 (pure-greedy, all ranges) — 10×3 CA-CA heatmap grid (GT / predicted-with-seeded-contacts / |residual|) at each protein's chosen k.



\$ contact_seeding_search_all_ranges.ipynb

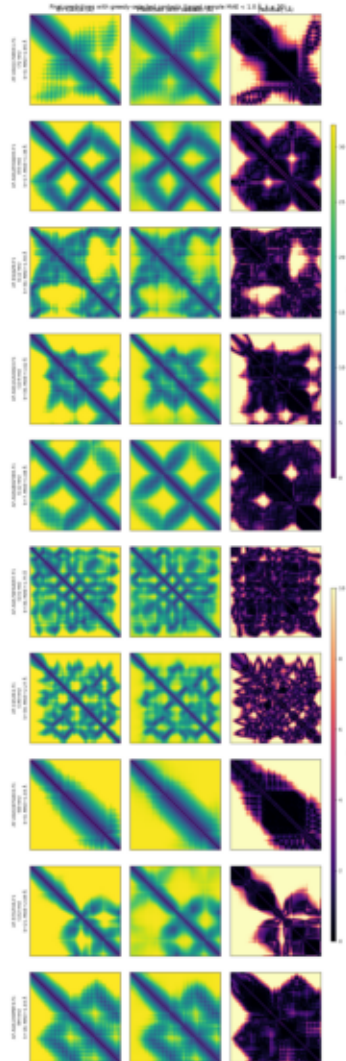
contact_search_all_ranges_trace.png

V3 (pure-greedy, all ranges) — MAE-vs-k curves; candidate pool opened to long + medium + short-range GT contacts. 4 / 10 proteins cross the 1.0 Å target; every protein improves vs V2.



contact_search_grid.png

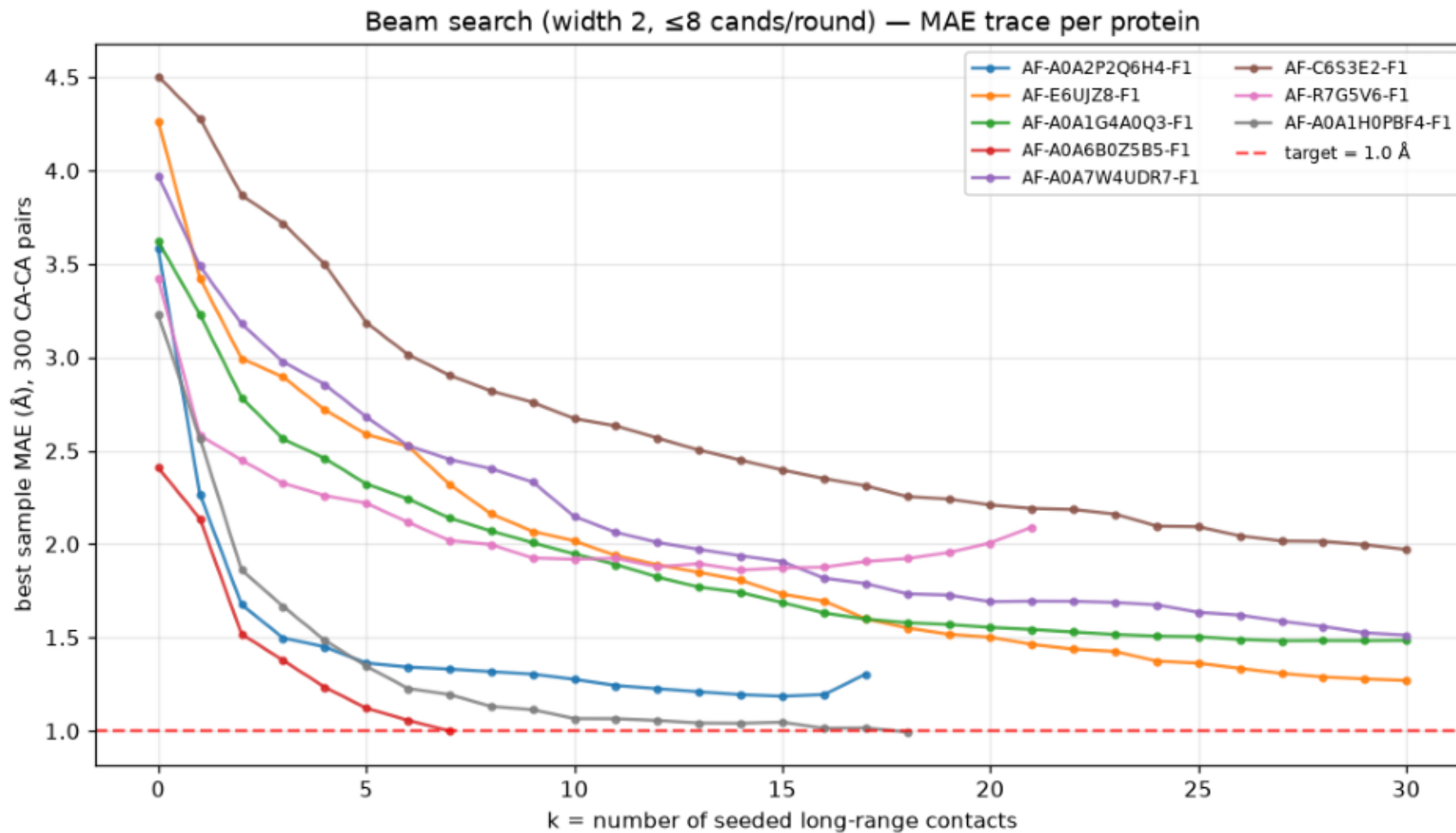
V2 (beam-2 long-range search) — 10×3 CA-CA heatmap grid (GT / predicted-with-seeded-contacts / |residual|) at each protein's chosen k.



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$ contact_seeding_search.ipynb
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contact_search_trace.png

V2 (beam-2 long-range search) — MAE-vs-k curves; one line per protein. Horizontal target at 1.0 Å. 2 / 10 proteins cross the threshold (AF-A0A6B0Z5B5-F1 at k=7, AF-A0A1H0PBF4-F1 at k=18).



pooled_scatter.png

V1 (zero-shot) — pooled expected vs GT CA-CA distance scatter across all 40,442 evaluable pairs from the 10 test proteins. Macro CA-CA MAE = 3.29 Å.

